MIL-P-62495A(AT)
30 May 1990
SUPERSEDING
(see 6.5)

MILITARY SPECIFICATION

POWER UNIT, HYDRAULIC

This specification is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hydraulic power unit which is a self-contained electrically driven assembly capable of supplying not less than 2 gallons per minute (gpm) of hydraulic fluid at 500 pounds per square inch gauge (psig) and not less than 1 gpm at 2000 psig. The hydraulic power unit shall be referred to by the term, "power unit" (see 6.1).

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. - The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS

FEDERAL

P-C-437	- Cleaning Compounds, High Pressure (Steam) Cleaner.
QQ-P-35	- Passivation Treatments for Corrosion Resistant Steel.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatment for Organic Coatings.

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MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-H-5606	- Hydraulic Fluid, Petroleum Eke: Aircraft, Missile and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-C-22750	- Coating Epoxy - Polyamide.
MIL-P-23377	Primer Coatings: Epoxy - Polyamide. Chemical and Solvent Resistant.
MIL-H-46170	- Hydraulic Fluid, Rust Inhabitant, Fire Resistant, Synthetic Hydrocarbon Base.
MIL-P-52192	- Primer Coating, Epoxy.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting Lead and Chromate Free.
MIL-P-53030	- Primer Coating, Epoxy. Water Reducible, Lead and Chromate Free.
MIL-H-83282	- Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft. Metric, NAT0 Code Number H-537.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

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MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461A{6)	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462(3)	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ARMY

12293172	-	Ramp Lock Installation.
12298252	-	Power Supply, Hydro-electrical.
12359539	-	Filter, EMI.

(Copies of drawings and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government drawings. The following document(a) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ASTM B633	-	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.

(Application for copies should be addressed to the ASTM 1916 Race Street. Philadelphia, PA 19103.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2-3 Order of precedence. In the event of a conflict between the text of this Specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulation unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish power units which shall be subjected to first article inspection (see 4.4). First article inspection samples. properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent power units delivered to the Government shall conform to these samples in all of their pertinent physical and performance attribute.

3.2 Materials. The materials selected shall be uniform in quality and free from defects and imperfections that might affect their serviceability. Supplier shall maintain traceability at incorporated material and provide physical or chemical certification (or both) of conformance to applicable material standards. Certifications shall precede or accompany shipment of each production lot (see 4.6.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.5).

3.3 Design and construction. The power unit shall conform to this specification and drawing 12298252 (see 4.6.2).

3.3.1 Interchangeability. All subassemblies and components having the same part number shall be functionally and dimensionally interchangeable (see 4.6.2).

3.3.2 Metals. All metals shall be compatible with the temperature and with the functional, service and storage conditions to which the components shall be exposed. The metals shall be suitably protected in accordance with 3.5.1. Magnesium shall not be used in any component (see 4.6.1).

3.3.3 Dissimilar metals. Compatibility of metals in contact shall be in accordance with MIL-STD-886 (see 4.6.1).

3.3.4 Hydraulic fluid. The power unit shall be compatible with fluids conforming to MIL-H-6083, MIL-H-5606, MIL-H-46170 and MIL-H-83282 (see 4.6.1).

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3.3.5 Insulation. Electrical components shall be insulated from case structure . The power unit shall not be grounded internally (ace 4.6.2).

3.3.6 High pressure filter. The pump discharge line shall include a 10 micron or finer filter (see 4 6.2).

3.3.7 Electric power. The pump and solenoid valve shall operate on 18 to 30 Volt direct current (Vdc) with a current draw of no more than 140 amps (A) to develop a pressure of 2,000 psig. Electrical connections shall be as shown on drawing 12298252 (see 4.6.1 and 4.6.2).

3.3 8 Weight. The power unit shall weigh no more than 55 pounds (lb) dry weight (see 4.6.2).

3.3.9 Protective coating.

3.3.9.1 Steel components. Exposed steel components shall be cadmium plated in accordance with QQ-P-416, type II, class 3. or zinc plated in accordance with ASTM B633, type II, class Fe/Zn 5, and exposed corrosion resisting steel components shall be passivated in accordance with QQ-P-35 (see 4.6.1).

3.3.9.2 Aluminum components. Aluminum components shall be treated in accordance with MIL-A-8625, type II, class 1 (see 4.6.1).

3.4 Performance. During the performance of the foregoing as well as environmental inspections, leaks and drips shall not be permissible. Weeps or seeps shall be allowable (see 6.3.2, 6.3.4, 6.3.6 and 6.3.8).

3.4.1 Fluid output. The power unit shall deliver no less than 2.0 gpm at 500 psig and no less than 1.0 gpm at 2,000 psig (see 4.6.3).

3.4.1.1 Relief valve. The relief valve shall start to open (crack) at 2,000 ± 100 psig. Full volume flow shall be provided at 2,250 ± 250 psig (see 4.6.3.1).

3.4 1.2 Leakage. The power unit under 500 psig pressure shall leak no more than 0.2 cubic inches of fluid in 10 seconds before and after meeting requirement 3.4.5. Under normal operation weeps and seeps shall be permissible and leaks or drips shall not be permissible (see 4.6.3.2 and 6.3).

3.4.2 Slope operation. The power unit shall be capable of operating on longitudinal and lateral slopes of ± 66 percent (%). When mounted in longitudinal or lateral directions, no fluid shall leak from the power unit (see 4.6.4).

3.4.3 Solennoid valve operation. When actuated by an 18 to 30 V dc signal, the solenoid valve shall reverse the flow between the outlet and discharge ports. The valve shall shift back to the original position when electrical power is removed (see 4.6.5).

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3.4.4 Electromagnetic interference (EMI). The power unit shall conform to MIL-STD-461A(6), requirements CE07 and RE05, except that the RE05 limits may be exceeded by no more than 35 decibels (db) in the frequency range of 2 Megahertz (MHz) to 300 MHz. In order to conform to this requirement, an EMI filter, part number 12359539, may be used (see 4.6.6).

3.4.5 Endurance. The hydraulic power unit shall operate the ramp or simulated ramp, satisfactorily for 1500 cycles from the fully extended to full retracted positions of the ramp. Total external leakage during the endurance teat shall be no greater than 15 percent of the reservoir fluid volume prior to the test. Fluid shall be added and recorded as necessary during endurance. Following this test the number of cycles completed in 5 minutes shall not be less than 80 percent of those at the beginning of the test (see 4.6.7).

3.4.6 Environmental conditions.

3.4.6.1 High temperature. After storage at 160 degrees Fahrenheit (°F) for a period of 48 hours, the power unit shall be capable of delivering no less than 0.4 gpm at 2,000 psig at an operating temperature of 160°F. After operation at 160°F the power unit shall not exhibit structural damage or deformation and shall meet the requirements of 3.4.1 through 3.4.3. The power unit shall also meet the performance requirements of 3.4.1 when operated intermittently (see 6.3.3) at 125°F for a period of 30 minutes (see 4.6-8.1).

3.4.6.2 Low temperature. After storage at -70°F, the power unit shall operate at -65°F and shll meet the performance requirement of 3.4.1 through 3.4.3 (see 4.6.8.2).

3.4.6.3 Humidity. After exposure to 10 cycles of 95% relative humidity at a temperature of 86°F, the power unit shall meet the performance requirements of 3.4.1 through 3.4.3 at the conclusion of the fifth and tenth cycle (see 4.6.9).

3.4.6.4 Steam and waterjet cleaning. The power unit shall meet the performance requirements of 3.4.1 through 3.4.3 after being subjected to the steam and waterjet cleaning specified in 4.6.10.

3.4.6.5 Shock. The power unit shall meet the requirements of 3.4.1 after application of the following shock loading (see 4.6.11).

a. Power unit mounting - As shown on drawing 12293172.
b. Area of shock application - At opposite face of chock mount from power unit.

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3.4.6.6 Vibration. The power unit shall meet the requirements of 3.4.1 after the following vibration (see 4.6.12].

a. Power unit mounting - As shown on drawing 12293172.
b. Area of vibration application - At opposite face of shock mount from power unit.
c. Level as shown in figure 1.

3.4.6.7 Fungus resistance. The unit shall meet requirements after being subjected to the fungus test for 90 days as described in MIL-F-13927, class 2, method B (ace 4.6.13).

3.4.6.8 Corrosion resistance. The unit shall meet requirements after exposure to the salt test in accordance with MIL-STD-202, test condition A of method 101D (see 4.6.14).

3.4.6.9 Sand and dust. The unit shall meet requirements after exposure to the sand and dust test in accordance with MIL-STD-202, method 110A (ace 4.6.15).

3.5 Finish. All exposed ferrous and aluminum surfaces shall be cleaned, treated, primed and finished as specified herein [ace 4.6.2).

3.5.1 Ferrous metals.

3.5.1.1 Cleaning and treatment.

a. Clean ferrous metals (Rockwell hardness less than C40) in accordance with the applicable method (1 through VI) specified in TT-C-490. Treat in accordance with type I or III of TT-C-490.

b. Clean ferrous metals (Rockwell hardness C40 through C47) in accordance with the applicable method (I through VI) specified in TT-C-490. Treat in accordance with type I or III of TT-C-490 (bake for hydrogen embrittlement relief as required).

c. Clean ferrous metals (Rockwell hardness C48 and over) in accordance with method I of TT-C-490. Acid solutions shall not be used for cleaning.

3.5.1.2 Priming. Apply a primer coating in accordance with MIL-P-53030, MIL-P-53022 or MIL-P-52192, composition optional. The dry film thickness of the primer shall be not lees than 1.0 mil.

3.5.1.3 Finishing. Apply an epoxy-polyamide coating conforming to type I of MIL-C-22750 or a modified high solids epoxy which meets the performance requirements of MIL-C-22750. The dry film thickness of the coating shall be not less than 1.0 mil. The color shall match color chip number 24533 (green) of FED-STD-595.

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3.5.2 Aluminum.

3.5.2.1 Treatment. Treat in accordance with class 1A of MIL-C-5541.

3.5.2.2 Priming. Apply a primer coating in accordance with MIL-P-53030, MIL-P-53022 or MIL-P-23377, composition optional. The dry film thickness of the primer shall be not less than 1.0 mil.

3.5.2.3 Finishing. Apply an epoxy-polyamide coating conforming to type I of MIL-C-22750 or a modified high solids epoxy which meets the performance requirements of MIL-C-22750. The dry film thickness of the coating shall be not less than 1.0 mil. The color shall match color chip nuber 24533 (green) of FED-STD-595.

3.6 Marking. The power unit shall be marked in accordance with MIL-STD-130 (ace 4.6.2).

3.7 Workmanship. Workmanship shall be such as to provide a power unit free from defects such as cracked, dented, loose, broken or corroded parts and components, burrs or sharp edges and scratched finish (see 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor-s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operation, is an acceptable practice to ascertain conformance to requirements; however, it does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

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4.3 Inspection conditions: Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ± 18°F.
b. Barometric pressure: 28.5 + 2 inches Hg.
- 3
c. Relative humidity: 35 ± 5%.
d. Voltage: 27.5 ± .5 V dc.

4.3.1 Test conditions. Test specimens shall be stabilized at room temperature for one hour before tests are begun.

4.3.1.1 Temperature extremes. The power unit shall be operated in a temperature range of 160 to -65°F.

4.3.1.2 Flow rate. The flow rate through the power unit shall vary from 2.6 to 1.0 gpm.

4.3.2 Mounting of sample. For performance tests (3.4.1 through 3.4.4) mounting and orientation of the sample will be as in the vehicle. The longest dimension will be the longitudinal dimension.

4.3.3 Test apparatus accuracy. Accuracy of test apparatus shall be as specified in MIL-STD-810.

4.4 First article inspection. First article inspections shall be performed on preproduction and initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply hydraulic power units that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection (PPI). Unless otherwise specified (see 6.2), the preproduction inspection emmple-shall consist of two power units. The sample units shall be subjected to the PPIs specified in table I.

4.4.2 Initial production inspection (IPI). Unless otherwise specified (see 6.2), the initial production sample shall-consist of three power units randomly selected from the first production lot manufactured. The sample units shall be subjected to the IPIs specified in table I.

4.4.3 First article inspection failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection, shall be evidence that all items already produced prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article inspection in completed to the satisfaction of the Government.

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4.5 Quality conformance inspection (QCI). QCI shall consist of the examinations of 4.5.2 and the acceptance tests of 4.5.3. Nonconformance with any of the specified requirements in sections 3 and 5 shall be cause for rejection.

4.5.1 Lot formation. A lot shall consist of all power units manufactured under the same condition during an identifiable production period with material traceability and submitted at the same time for acceptance.

4.5.1.1 Sample size. Unless otherwise specified (see 6.2) sample size for quality conformance examination and acceptance tests shall be determined in accordance with sampling inspection table II. For definitions of terms used with sampling inspection see 6.3.8.

TABLE I. Classification of inspection.

Title	Requirement	Inspection	First article		QCI	
			PPI	IPI	Examination	Acceptance tests
Materials and construction	3.2, 3.2.1, 3.3.2 thru 3.3.4, 3.3.7 and 3.3.9	4.6.1	X	X		
Defects (see 4.5.2 and table II)	3.3, 3.3.1, 3.3.5 thru 3.3.8, 3.5 thru 3.7	4.6.2	X	X	X	
Fluid output	3.4.1	4.6.3	X	X		X 1/
Relief valve	3.4.1.1	4.6.3.1	X	X		X
Leakage	3.4.1.2	4.6.3.2	X	X		X
Slope operation	3.4.2	4.6.4	X	X		X
Solenoid valve operation	3.4.3	4.6.5	X	X		X
EMI	3.4.4	4.6.6	X	X		X
Endurance	3.4.5	4.6.7	X	X		X
High temperature	3.4.6.1	4.6.8.1	X	X		
Low temperature	3.4.6.2	4.6.8.2	X	X		
Humidity	3.4.6.3	4.6.9	X	X		
Steam and water-jet cleaning	3.4.6.4	4.6.10	X	X		
Shock	3.4.6.5	4.6.11	X	X		
Vibration	3.4.6.6	4.6.12	X	X		
Fungus resistance	3.4.6.7	4.6.13	X	X		
Corrosion resistance	3.4.6.8	4.6.14	X	X		
Sand and dust	3.4.6.9	4.6.15	X	X		

1/ 100% testing for fluid output performance.

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4.5.2 Examinations. The sample units shall be examined for the defects specified in table III.

4.5.3 Acceptance tests. The sample units (selected in accordance with table II) shall be subjected to the acceptance tests specified in table I, except fluid output requirement. All power units shall be tested for fluid output performance.

4.5.4 Failure disposition. Any examination defect or acceptance test failure shall be cause for the Government to refuse acceptance of the production quantity represented by the lot size. Any deficiency found during or as a result of QCI shall be evidence that all items represented by the lot size are similarly deficient unless contrary evidence satisfactory to the Government is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor.

TABLE II. Sample inspections.

Sampling lot size	Sample size for Majors	Sample size for Minors	Acceptance Tests
2 to 8	*	3	*
9 to 15	13	3	13
16 to 25	13	3	13
26 to 50	13	5	13
51 to 90	13	6	13
91 to 150	13	7	13
151 to 280	20	10	20
281 to 500	29	11	29
501 to 1200	34	15	34
1,201 to 3200	42	18	42
3,201 to 10,000	50	22	50
10,001 to 35,000	60	29	60
35,001 to 150,000	74	29	74
150,001 to 500,000	90	29	90
500,001 and over	102	29	102

* Indicates entire lot must be inspected.
NOTE: The acceptance number in all cases is zero.

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TABLE III. Classification of defects.

Category	Defects	Method of examination
Critical	None	
Major		
101	Nonconformance in design and construction (see 3.3).	Visual
102	Dimensions affecting interchangeability out of tolerance (see 3.3).	SIE 2/
103	Faulty workmanship affecting performance (see 3.7).	Visual
Minor		
201	Dimensions not affecting interchangeability out of tolerance (see 3.3).	SIE
202	Weight not as specified (see 3.3.8).	SIE
203	Improper finish (see 3.5).	Visual
204	Improper marking (see 3.6).	Visual
205	Faulty workmanship affecting appearance (see 3.7).	Visual

2/ SIE - Standard Inspection Equipment.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2, 3.2.1, 3.3.2 through 3.3.4, 3.3.7 and 3.3.9 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, teat reports, and rating data.

4.6.2 Defects. Conformance to 3.3, 3.3.1, 3.3.5 through 3.3.8 and 3.5 through 3.7 shall be determined by examination for the defects listed in table I. Examination shall be visual, or by measurement with standard inspection equipment.

4.6.3 Fluid output. To determine conformance to 3.4.1, when filled with fluid conforming to 3.3.4 and energized with 24 V dc, the power unit shall provide no less than 1.0 gpm at 1,000 psig and no less than 2.0 gpm at 500 psig.

4.6.3.1 Relief valve. To determine conformance to 3.4.1.1. verify that the relief valve in the pump discharge line shall begin to open at 2,000 ± 100 psig.

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4.6.3.2 Leakage. To determine conformance to 3.4.1.2, during the performance and environmental tests verify that there are no leaks or drips. Weeps or seeps are allowable. With 500 psig pressure applied to the "p", pressure port side, of the power unit when the solenoid is de-energized, fluid leakage shall not exceed 0.2 cubic inches in 10 seconds before and after the endurance test.

4.6.4 Slope operation. To determine conformance to 3.4.2, it shall be verified that flow outputs and pressures specified in 3.4.1 are maintained when the unit is inclined longitudinally at ± 66% and laterally at ± 66% from the horizontal.

4.6.5 Solenoid valve operation. To determine conformance to 3.4.3, verify that the solenoid valve, when energized with 18 to 30 V dc, reverses the direction of flow between tank port and pressure port at both 500 and 2,000 psig, and that the valve shifts back when voltage is removed.

4.6.6 Electromagnetic interference. To determine conformance to 3.4.4, the power units shall be tested in accordance with MIL-STD-462(3). If the EMI filter 12359539 is required, the power unit equipped with filter shall be installed in a vehicle and tested as specified in MIL-STD-461A(6). Unless otherwise specified. vehicle testing shall be the responsibility of the procuring activity (see 6.2).

4.6.7 Endurance. To determine conformance to 3.4.5, the hydraulic power unit shall be operated intermittently for 1500 cycles from the fully extended to fully retracted positions with 5 minutes between cycles. During the test the unit shall operate the ramp or a simulated ramp of equivalent weight-

4.6.7.1 Endurance details. Intermittent operation is defined as operation for 5 minutes followed by nonoperation for 5 minutes. A cycle is defined as operation of the unit from fully retracted position to fully extended Position and return.

4.6.8 Environmental conditions. The environmental tests of 4.6.8.1 through 4.6.12 shall be conducted to determine conformance to 3.4.6. The performance tests of 4.6.3 through 4.6.6 shall be performed on all samples which have satisfactorily undergone the environmental tests. Verify that there is no degradation of performance.

4.6.8.1 High temperature. To determine conformance to 3.4.6.1, the hydraulic power unit shall be tested according to MIL-STD-810D, Method 501.2, procedures I and II within the following limits and guidelines. The storage temperature shall be 160°F, held constant for 48 hours. Following storage the power unit shall be operated intermittently for 1,000 cycles in accordance with 4.6.3. at a stabilized 160°F.

4.6.8.2 Low temperature. To determine conformance to 3.4.6.2, the hydraulic power unit shall be tested per MIL-STD-810D, Method 502.2, Procedures I and II within the following limits and guidelines. The storage temperature shall be -70°F, held constant for 48 hours. Following storage the power unit shall be operated intermittently for 100 cycles in accordance with 4.6.3, at a stabilized -65°F.

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4.6.9 Humidity. To determine conformance to 3.4.6.3, the hydraulic power unit shall be tested in accordance with MIL-STD-810D, Method 507.2, Procedure III. Near the end of the fifth and tenth cycles with humidity remaining at 95%, operate hydraulic power unit and record results (ace figure 1).

4.6.10 Steam and waterjet cleaning. To determine conformance to 3.4.6.4, the hydraulic power unit shall be steam cleaned, using high pressure steam with cleaner conforming to P-C-437. After steam cleaning, the unit shall be waterjet washed. The method for cleaning shall be as follows: The jet shall be applied perpendicular to the surface being cleaned, at a distance of not more than one foot for steam cleaning and not more than five feet for waterjet cleaning. The jet pressure shall be 100 to 110 psig and the jet shall be moved at such a rate that the exposed surface of the power unit is subjected to the jet at a rate not to exceed one square foot per minute. After testing, the power unit shall be tested as specified in 4.6.3 through 4.6.5.

4.6.11 Shock. To determine conformance to 3.4.6.5. the hydraulic power unit shall be tested in accordance with MIL-STD-810D, Method 516.3, procedure I, figure 516-3-4, 40 g peak value, 11 millisecond duration. The shock shall be applied to the face of foot mount opposite from the unit. Following the teat the unit shall have no deformed, cracked, or broken parts (see figure 2).

4.6.12 Vibration. To determine conformance to 3.4.6.6, the hydraulic power unit shall be tested in accordance with MIL-STD-810D, Method 514.3, Category 8, table 514.3-VII, and figure 514.3-35. Additional test conditions shall be as follows:

a. Sinusoidal vibration tests shall be conducted within the limits of figure 4.

b. Maximum load along the longitudinal axis shall be 10 g.

c. During the test, operation of the unit is unnecessary.

d. The unit shall be vibrated along each axis in accordance with the test levels, frequency range and times from figures 3 and 4.

e. Following the vibration tests the unit shall have no deformed, cracked or broken parts.

4.6.13 Fungus resistance. To determine conformance to 3.4.6.7, the hydraulic power unit shall be tested in accordance with MIL-F-13927, class 2, method B except that the test period shall be a continuous 90 days. Upon removal from the cabinet, examination shall be accomplished by means of removable plates or plugs, without disassembly and without disturbing any fungus growths that have developed. Upon completion of the examination and replacement of access port covers, the unit shall be operated intermittently for one hour. in accordance with 4.6.1.

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4.6.14 Corrosion resistance. To determine conformance to 3.4.6.8, the hydraulic power unit shall be tested for resistance to corrosion in accordance with MIL-STD-202, test condition A of method 101D. Following the test, unit shall be operated intermittently for one hour in accordance with 4.6.3.

4.6.15 Sand and dust. To determine conformance to 3.4.6.9, the hydraulic power unit shall be subjected to the sand and dust test of MIL-STD-202, method 110A. During step 1 and 3 of the procedure the unit shall operate intermittently at full stroke and at a no load condition. Intermittent operation is described in 4.6.7.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The power unit manufactured in accordance with this specification is intended for ramp operation on vehicles of the Fighting Vehicle System.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required. the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. If first article is required (see 3.1).
d. If responsibility for inspection and place of inspection is other than as specified (see 4.1).
e. If inspection conditions shall be other than as specified (see 4.3).
f. If preproduction sample size is other than as specified (see 4.4.1).
g. If initial production sample size is other than as specified (see 4.4.2).
h. If quality conformance examination and acceptance test sample size is other than as specified (see 4.5.1.1).
i. If responsibility for vehicle testing to verify EMI requirement shall be other than as specified (see 4.6.6).
j. Selection of applicable level and packaging requirements (see 5.1).

6.3 Definition.

6.3.1 Cycle. A cycle is defined as operation of the unit from fully retracted position to fully extended position and return.

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6.3.2 Drip. Any evidence of fluid beyond a seal or splitline where droplets for-m and fall.

6.3.3 Intermittent operation. Intermittent operation is defined as operation for 30 seconds and non-operation for 2-1/2 minutes.

6.3.4 Leak. Any evidence of fluid beyond a seal or splitline that results in the formation of a droplet.

6.3.5 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.7).

6.3.6 Seep. Any evidence of fluid beyond a seal or splitline that does not result in the formation of a droplet.

6.3.7 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid. liquid, semisolid, or contained gaseous material resulting from industrial, commercial. mining and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage. or solid or dissolved material in irrigation return flows, or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations. section 23.402).

6.3.8 Terms used with sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirement. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other theses, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product; or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a chip, aircraft, tank, missile. or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

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d. Defective. A defective is a unit of product which contains one or more defects.

e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 6.3.81). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions. and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot" or "inspection batch", i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment. etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

1. Presentation of lots or batches. The formation of the lots or batches, lot or batch size. and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

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n. Sample A sample consists of one or more units of product drawn from a lot or batch. the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.3.9 Weep. Any evidence of fluid beyond a seal or splitline.

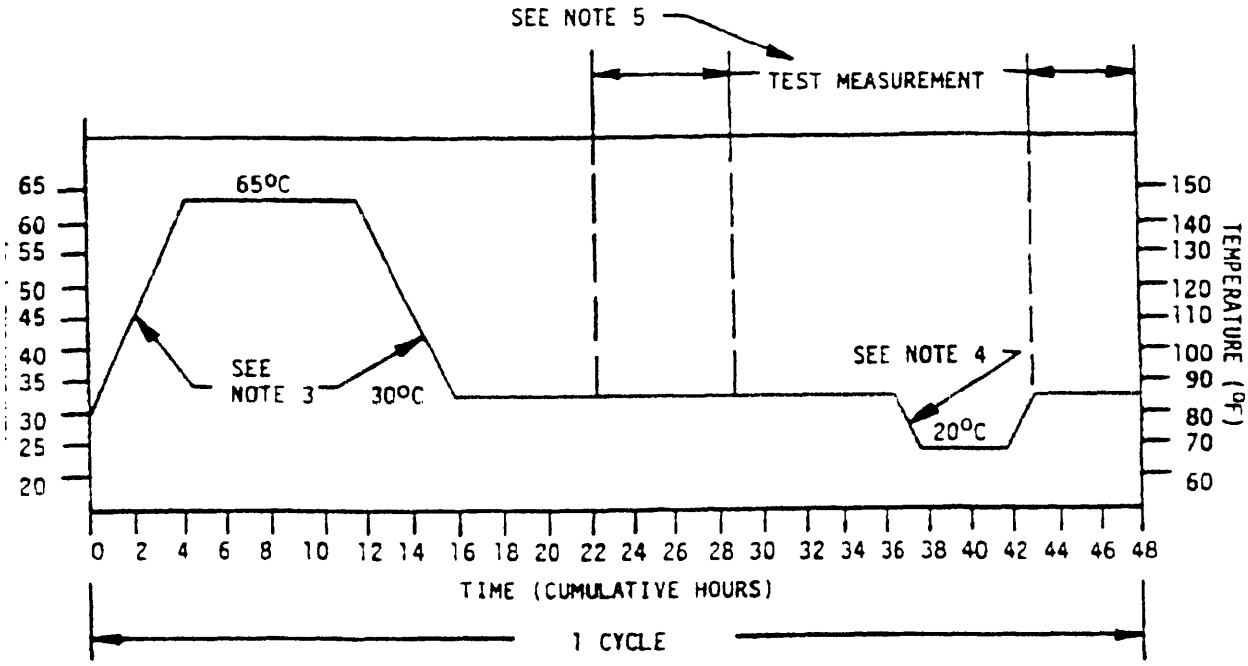
6.4 Subject term (key word) listing.
Electromagnetic interference
Filter, high pressure
Finish/surface treatment
Fluid output
High temperature
Humidity
Hydraulic fluid
Low temperature
Marking
Protective coatings
Recovered materials
Shock
Slope operation
Solenoid valve
Vibration

6.5 Supersession data. This specification supersedes MIL-P-62495, dated 17 December 1984, and incorporates the latest requirements in specification drawing 12292153. dated 19 November 1986.

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 AMC policy on AQLs and LTPDs. This specification is certified to be in accordance with current Army Materiel Command (MC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defective) from military specifications.

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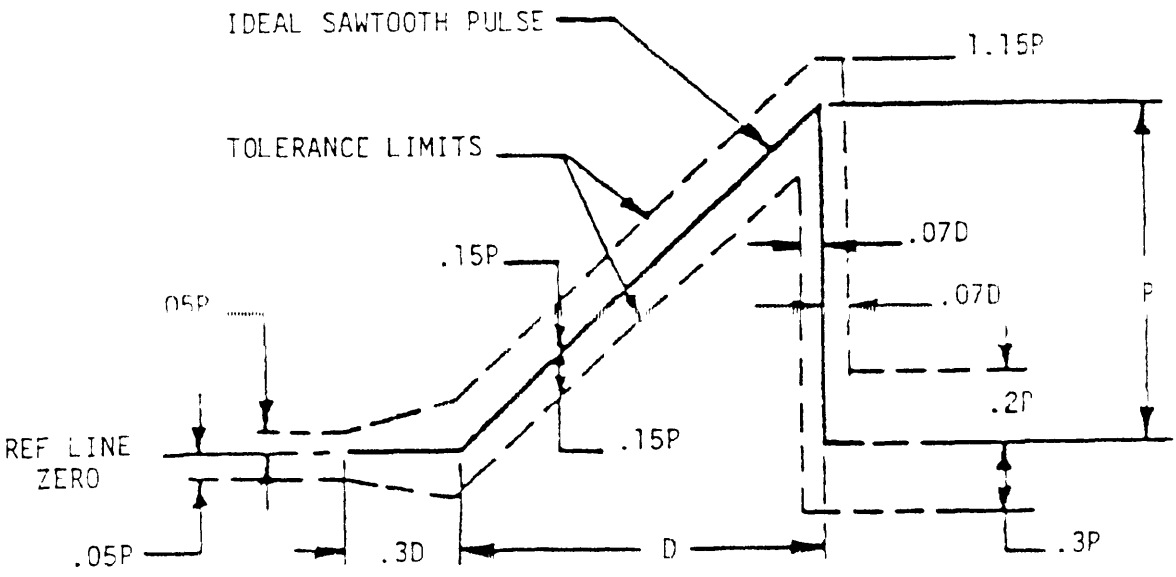


NOTES:

1. Tolerance during temperature change shall be not greater than 3°C (5°F).
2. Relative humidity shall be maintained at 94 ± 4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 30 and 65°C (86 and 149°F) shall be not less than 8°C (14.4°F) per hour.
4. The temperature increase in this portion of the curve shall be not less than 10°C (18°F).
5. Test measurements shall be taken only at the period specified in the applicable equipment or system specification.

FIGURE 1. Humidity Cycle.

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P = Peak value = 40 g
D = Nominal duration = 11 ms

NOTE. THE OSCILLOGRAM SHALL INCLUDE A TIME ABOUT 3D LONG WITH A PULSE LOCATED APPROXIMATELY IN THE CENTER. THE PEAK ACCELERATION MAGNITUDE OF THE SAWTOOTH PULSE IS P AND IT'S DURATION IS D. THE MEASURED ACCELERATION PULSE SHALL BE CONTAINED BETWEEN THE BROKEN LINE BOUNDARIES AND THE MEASURED VELOCITY CHANGE (WHICH MAY BE OBTAINED BY INTEGRATION OF THE ACCELERATION PULSE) SHALL BE WITHIN THE LIMITS OF $V_i \pm 0.1\ V_i$ WHERE V_i IS THE VELOCITY-CHANGE ASSOCIATED WITH THE IDEAL PULSE WHICH EQUALS 0.5 PD. THE INTEGRATION TO DETERMINE VELOCITY CHANGE SHALL EXTEND FROM 0.4D BEFORE THE PULSE TO 0.1D) AFTER THE PULSE.

FIGURE 2. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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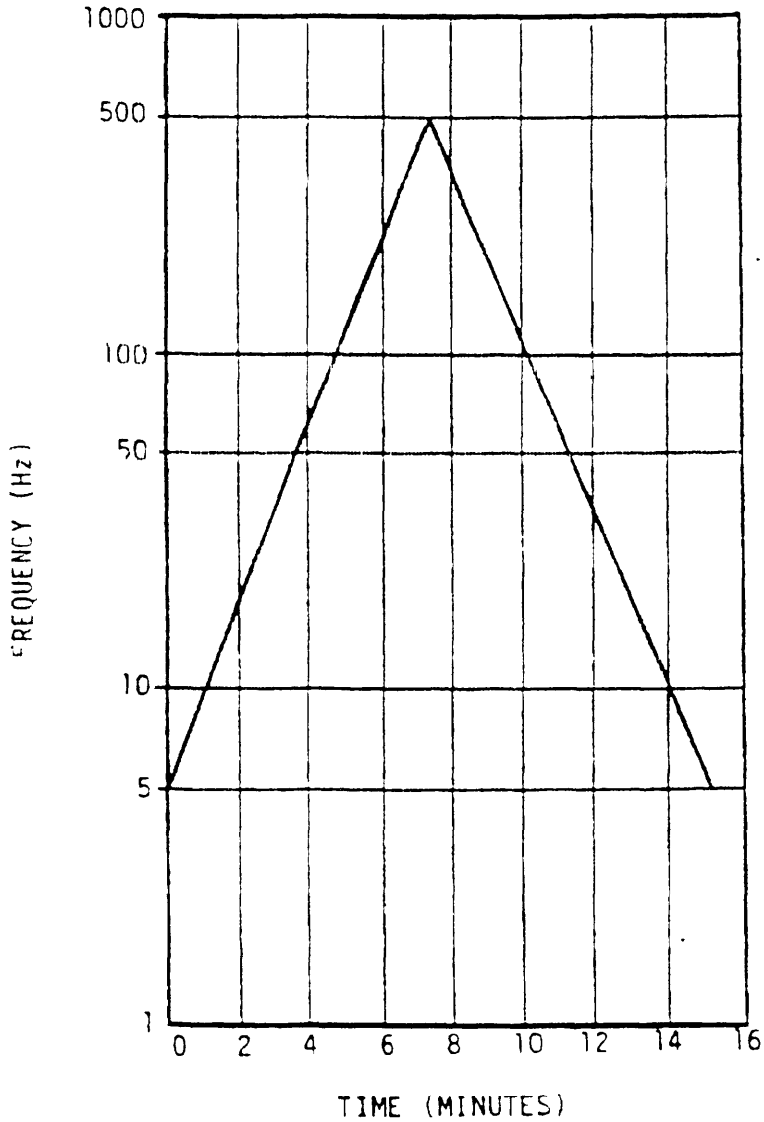
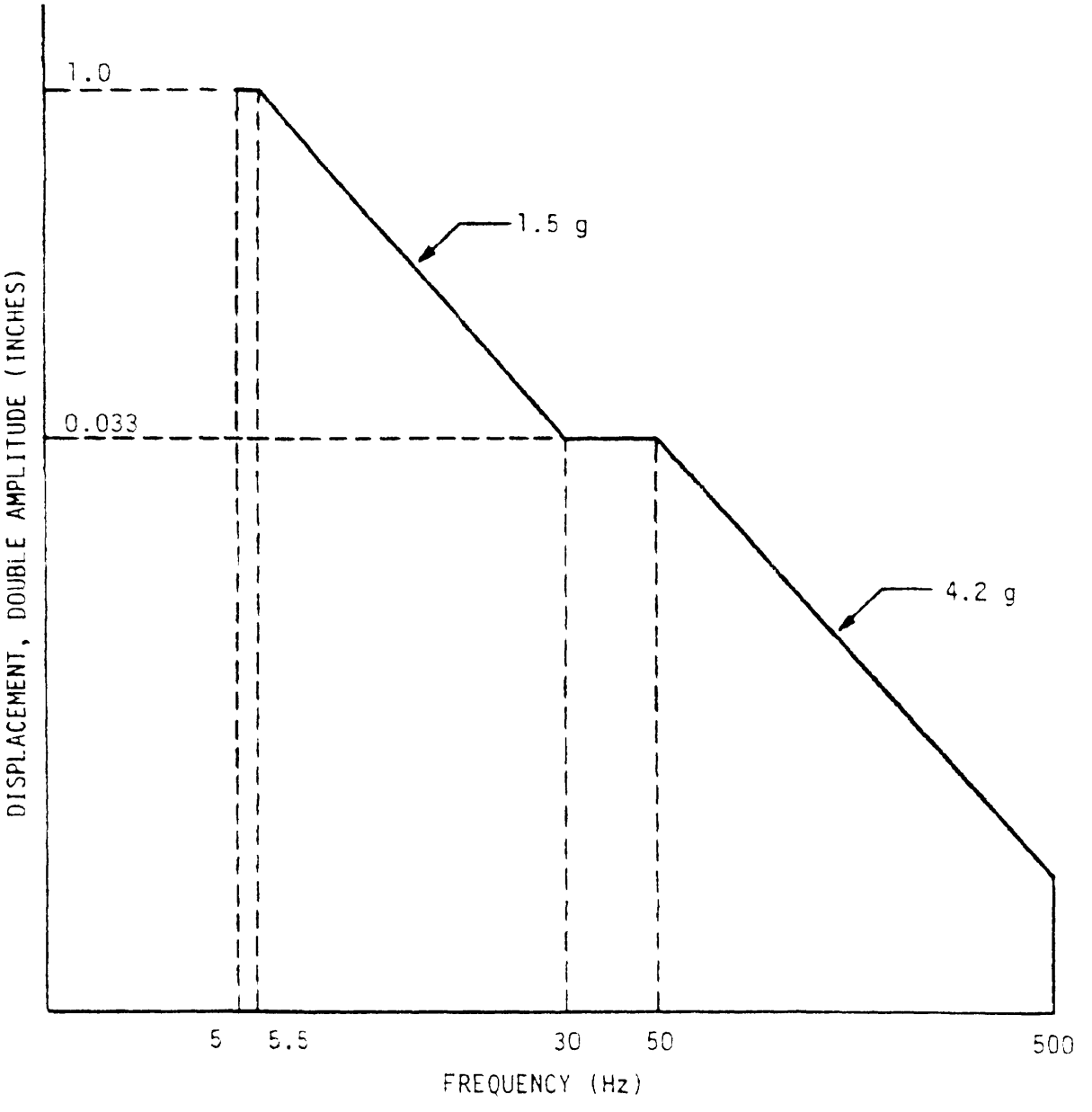


FIGURE 2. Logarithmic sweep.

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NOTES:

1. Acceleration levels: ± 9 g.
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 4. Vibration Test Curve.

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